TED(21)-5123C **REVISION 2021**

MODEL QUESTION PAPER

PROCESS CONTROL AND INSTRUMENTATION SET-1

Time : 3 Hours Max.Marks : 75

Part A

I. Answer **all** questions in one word or one sentence.

1. Define resolution.
2. Recall span and range.
3. List two level indicating devices.
4. Give one instrument used to measure viscosity.
5. Define dew point.
6. Identify the dielectric strength.
7. Recall bulk.
8. List 2 types of controllers.
9. Recall DCS. (9*1 =9)

Part B

II. Answer any ***eight*** questions from the following, Each question carries 3 marks.

1. Write a note on importance of instrumentation in industries.
2. Illustrate the working principle of float level indicator.
3. Explain briefly the working of ultrasonic flow meter.
4. List any four temperature measuring devices.
5. Explain the working of redwood viscometer.
6. List the optical properties measured in paper .
7. Explain the mechanism of solenoid valve
8. Construct the block diagram of control system.
9. Explain briefly PID controller
10. Discuss briefly emergency shutdown system. (8*3 =24)

Part C

Answer **all** questions. Each question carries 7 marks.

III. Discuss briefly the elements of measuring instruments.

OR

IV. Explain the on and offline measurements.

V. Write a note on (i) fidelity (ii) lag (iii)dynamic error.

OR

VI. Discuss transducers and their function .

VII. Explain the principle of working of bellows .

OR

VIII. Describe the method of pH measurement.

IX. Explain the electrical properties of paper.

OR

X. Describe the method of determination of (i) tear strength (ii) bending stiffness.

XI. Explain the method of determination of (i) caliper (ii) porosity.

OR

XII. Explain the water absorptiveness of paper.

XIII. Explain Programmable logic controllers.

OR

XIV. Explain the components of SCADA. (6*7 = 42)

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MODEL QUESTION PAPER 1

PROCESS CONTROL AND INSTRUMENTATION SET-2

Time : 3 Hours Max.Marks : 75

Part A

I. Answer **all** questions in one word or one sentence.

10. Define reproducibility
11. Recall dead zone.
12. Give two flow measuring devices.
13. Identify the instrument used to measure specific gravity.
14. Define kinematic viscosity.
15. List the optical properties of paper.
16. Recall folding endurance.
17. Give one advantage of PI controller over proportional controller.
18. Recall SCADA. (9*1 =9)

Part B

II. Answer any ***eight*** questions from the following, Each question carries 3 marks.

11. Write a note on dynamic characteristics of measuring instruments.
12. Illustrate the working principle of float level indicator.
13. Explain briefly the working of pyrometer.
14. List any four pressure measuring devices.
15. Explain the working of saybolt viscometer.
16. List the electrical properties measured in paper .
17. Sketch the standard instrumentation symbols for a) flow control valve b) solenoid valve c) pressure relief valve.
18. Compare negative and positive feedback control system.
19. Explain briefly RTU
20. State the disadvantages of distributed control system (DCS). (8*3 =24)

Part C

Answer **all** questions. Each question carries 7 marks.

III. Discuss transducers and their classification.

OR

IV. Explain an instrumentation block diagram briefly.

V. Write a note on (i) repeatability (ii) stability (iii) linearity.

OR

VI. Discuss the functions of measuring elements .

VII. Explain the following types of moisture measurement (i) Infrared type (ii)Capacitance type.

OR

VIII. Describe two methods of humidity measurement.

IX. Explain the resistance properties of paper.

OR

X. Describe the method of determination of strength properties of paper like tensile and bursting strength.

XI. Explain the method of determination of (i) grammage (ii) bulk.

OR

XII. Explain the air/gas permeability of paper.

XIII. Prepare a P&ID to show the major control schemes in a single effect evaporator.

OR

XIV. Explain the mechanism and principle of pneumatic control valve. (6*7 = 42)